MAUNA

2026-2027

ADVANCED GPS PERFORMANCE ANALYSIS – HAMBOURG

Club report — MAUNA data only

Mon, 05/10/26 · 2026-09-22 → 2026-10-05

01 Highest injury risk (50.1) with severe accel/decel imbalance (2.06:1) and declining max speed (7.6→7.2 km/h). Low subjective soreness (10-30) masks underlying neuromuscular fatigue.

02 Elevated risk (35.3) despite low mechanical load. DSL 75.9 m (50% below squad mean) indicates underload. Risk driven by fitness gap, not overtraining.

03 Highest HMLD exposure (10,078 m) with low injury risk (17.6). Balanced accel/decel (1.1:1) and stable max speed. Reference profile for squad capacity.

LOAD GPS WELLNESS SLEEP BIOMARKERS

HAMBOURG · PHYSICAL DATA

CONTENTS Hambourg

Advanced GPS Performance Analysis – Hambourg

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03 · GPS Performance Derivatives & Injury Risk Analysis §1

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STAFF READ

2026-09-22 → 2026-10-05

Technical GPS derivative analysis identifying mechanical load imbalances and neuromuscular decline patterns linked to elevated injury risk in key players.

SIGNALS

INFO

Highest injury risk (50.1) with severe accel/decel imbalance (2.06:1) and declining max speed (7.6→7.2 km/h). Low subjective soreness (10-30) masks underlying neuromuscular fatigue.

Albert Sambi Lokonga · gps · wellness

INFO

Elevated risk (35.3) despite low mechanical load. DSL 75.9 m (50% below squad mean) indicates underload. Risk driven by fitness gap, not overtraining.

Yussuf Poulsen · gps · load

INFO

Highest HMLD exposure (10,078 m) with low injury risk (17.6). Balanced accel/decel (1.1:1) and stable max speed. Reference profile for squad capacity.

Miro Muheim · gps · load

CANVAS SECTIONS

Squad Mechanical Load & Intensity Profile	TEAM
Intensity Derivatives & Mechanical Efficiency	TEAM
Case Study: Albert Sambi Lokonga	TEAM
Case Study: Yussuf Poulsen	TEAM
Case Study: Miro Muheim	TEAM
Cross-Domain Analysis	TEAM
Technical Recommendations	TEAM

GPS PERFORMANCE DERIVATIVES & INJURY RISK ANALYSIS

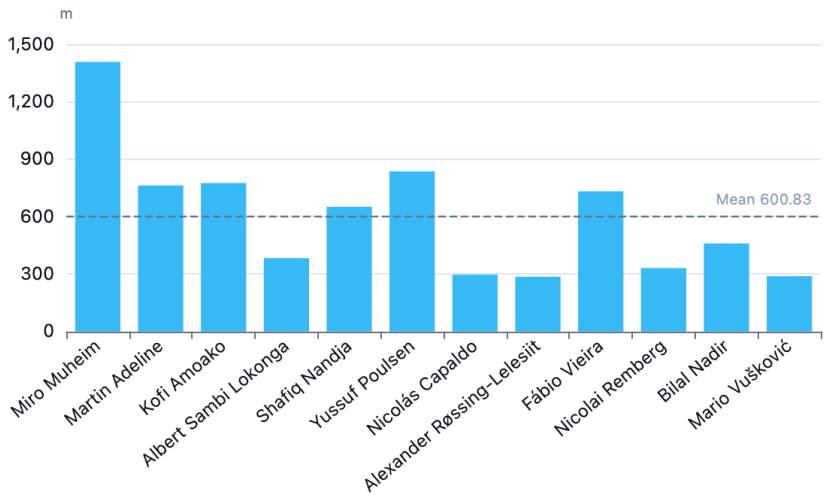
2026-09-22 → 2026-10-05

- Advanced technical analysis of GPS-derived performance metrics over 14 days (22 Sep – 5 Oct 2026). Focus on mechanical load patterns, intensity derivatives, and their relationship with injury risk and performance decline.

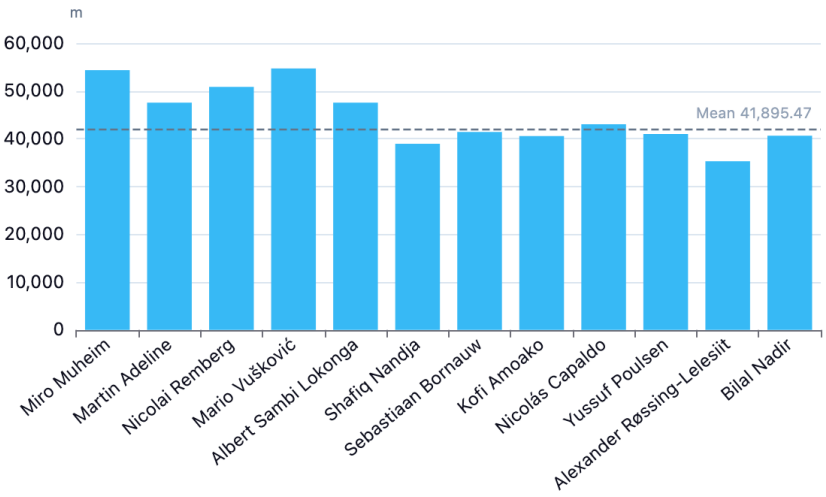
SQUAD MECHANICAL LOAD & INTENSITY PROFILE

2026-09-22 → 2026-10-05

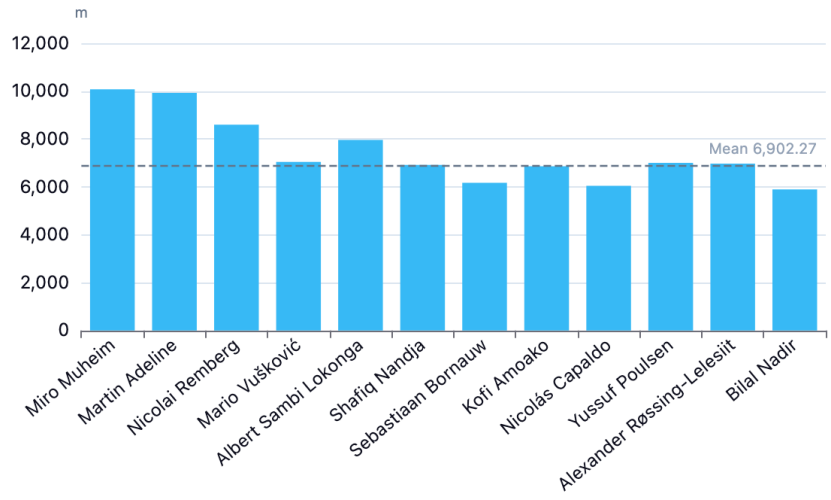
SPRINT DISTANCE VOLUME – TOP 12 PLAYERS (M)



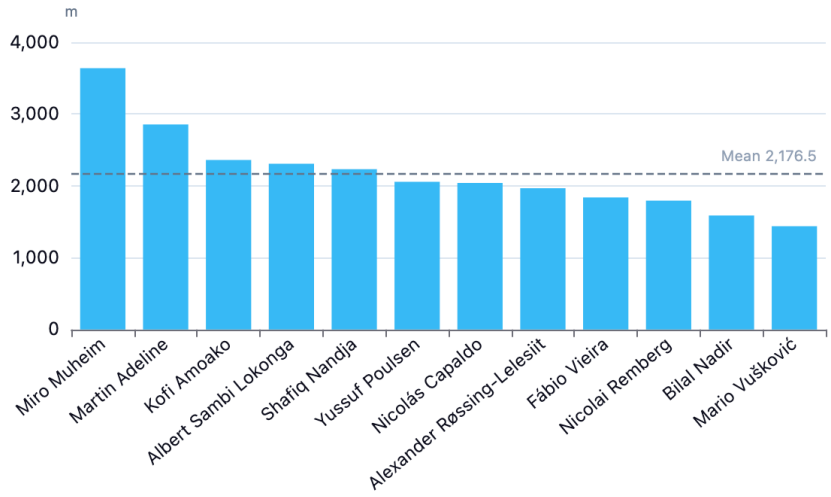
TOTAL DISTANCE COVERED – TOP 12 PLAYERS (M)



HIGH METABOLIC LOAD DISTANCE (HMLD) – TOP 12 (M)



HIGH-SPEED RUNNING VOLUME – TOP 12 PLAYERS (M)



SQUAD GPS PERFORMANCE SUMMARY (PERIOD TOTALS & AVERAGES)

high medium low vs squad (terciles)

Player	N	Distance total (sum)	HMLD (sum)	High speed running (absolute) (sum)	DSL (mean)	Max speed (max)	Performance Score Mean
Alexander Røssing-Lelesiit	12	35.3	7	2	188	8.8	52.9
Mario Vušković	12	54.7	7	1.4	181.4	9.2	51.1
Bilal Nadir	8	40.6	5.9	1.6	145	9.2	–
Bakery Jatta	12	24.4	3.9	1.2	167.9	8.7	55.2
Yussuf Poulsen	13	41	7	2.1	75.9	9	50.9
Kofi Amoako	11	40.5	6.9	2.4	219.6	9.4	–
Miro Muheim	13	54.3	10.1	3.6	191.2	9	49.8
Albert Sambi Lokonga	13	47.6	8	2.3	215.9	8.4	50.5
Fábio Vieira	11	37.8	5.9	1.8	189.2	8.4	49.7
Jordan Torunarigha	13	30.4	4.3	1.3	145.9	9.1	48.9
Nicolás Capaldo	13	43	6	2	178.3	9.5	50.5
Nicolai Remberg	13	50.9	8.6	1.8	352	9.2	69.3
Martin Adeline	5	47.5	9.9	2.9	386.4	9.3	–
Sebastiaan Bornauw	6	41.4	6.2	1.3	195.1	9.7	–
Terem Moffi	12	14.9	1.6	781	79.9	9	–
Shafiq Nandja	11	38.9	6.9	2.2	168.1	10.2	–
Albert Grønbæk	12	9.7	492	36	139.3	6.4	54.2
Otto Stange	12	7.8	1.6	701	108	8.8	51.2
David Möller Wolfe	4	–	–	–	–	–	–
Patson Daka	9	–	–	–	–	–	–
Sebastiaan Bornauw	5	–	–	–	–	–	–
Warmed Omari	11	–	–	–	–	–	50.9
Fabio Baldé	10	–	–	–	–	–	51
Jean-Luc Dompé	10	–	–	–	–	–	51
Damion Downs	10	–	–	–	–	–	51
Daniel Elfadli	10	–	–	–	–	–	51
Robert Glatzel	10	–	–	–	–	–	51
Giorgi Gocholeishvili	10	–	–	–	–	–	51
Noah Katterbach	10	–	–	–	–	–	51
Ransford Königsdörffer	10	–	–	–	–	–	51
Omar Megeed	10	–	–	–	–	–	51
William Mikelbrencis	10	–	–	–	–	–	51
Philip Otele	10	–	–	–	–	–	51
Immanuël Pherai	10	–	–	–	–	–	50.5
Rayan Philippe	10	–	–	–	–	–	51
Emir Sahiti	10	–	–	–	–	–	51

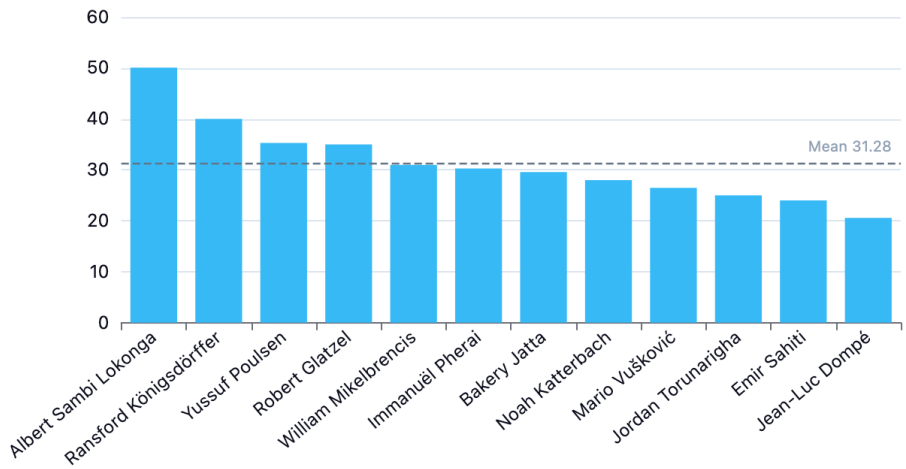
Player	N	Distance total (sum)	HMLD (sum)	High speed running (absolute) (sum)	DSL (mean)	Max speed (max)	Performance Score Mean
Luka Vušković	10	—	—	—	—	—	**51**

- Muheim leads HMLD exposure (10,078 m) with high sprint volume (1,408 m). Adeline shows extreme per-session intensity (mean 1,985 m HMLD/session, max 3,015 m). Sambi Lokonga combines moderate volume with elevated injury risk (mean 50.1). Poulsen shows high injury risk (35.3) despite lower HMLD, suggesting recovery deficit.

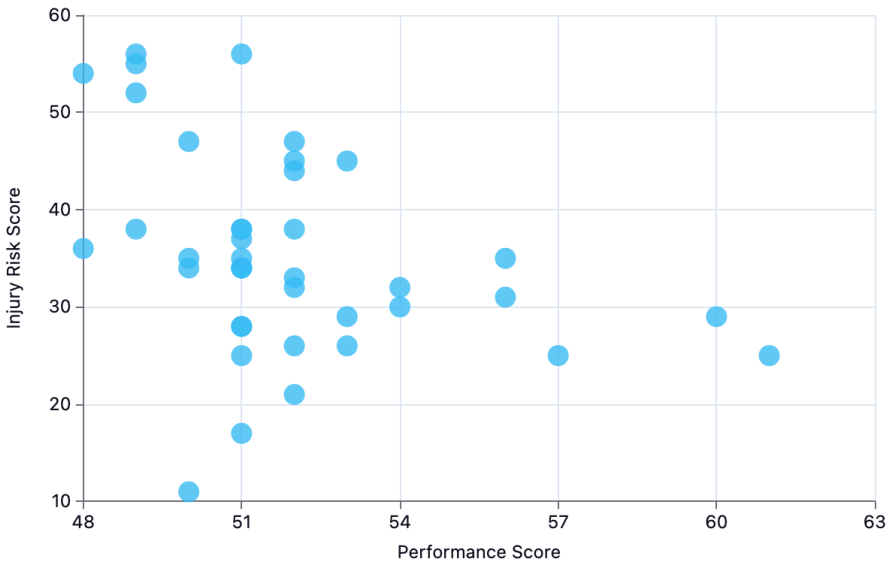
INTENSITY DERIVATIVES & MECHANICAL EFFICIENCY

2026-09-22 → 2026-10-05

MEAN INJURY RISK SCORE – TOP 12 PLAYERS



PERFORMANCE SCORE VS INJURY RISK – DAILY EVOLUTION



- Sambi Lokonga: injury risk 50.1 (highest in squad) with moderate performance 50.5. Poulsen: risk 35.3 despite lower HMLD volume, indicating insufficient recovery capacity. Königsdörffer & Glatzel: elevated risk without GPS data suggests non-training load factors. Correlation analysis: max speed vs injury r = -0.02 (n=45), body soreness vs performance r = -0.15 (n=56).

CASE STUDY: ALBERT SAMBI LOKONGA – HIGH INJURY RISK PROFILE

2026-09-22 → 2026-10-05

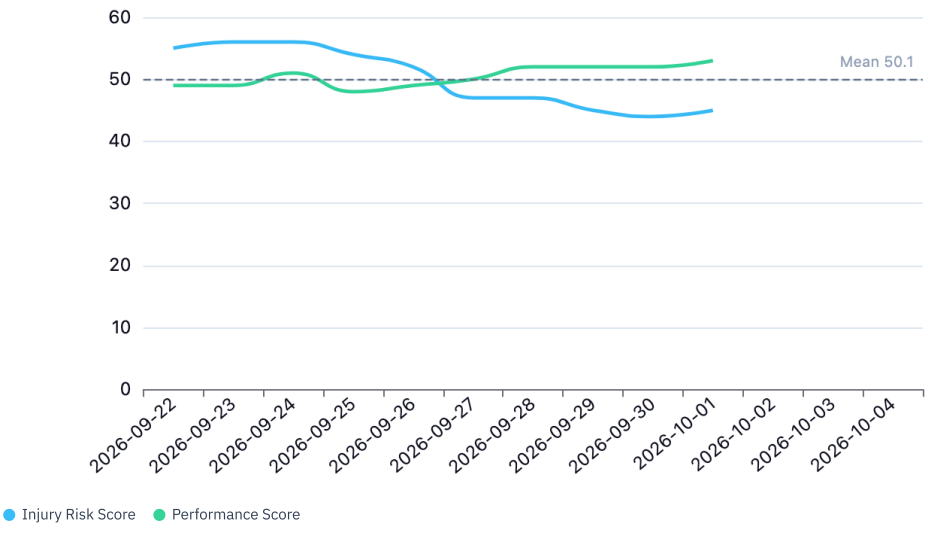
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Albert Sambi Lokonga
CENTRAL MIDFIELDER

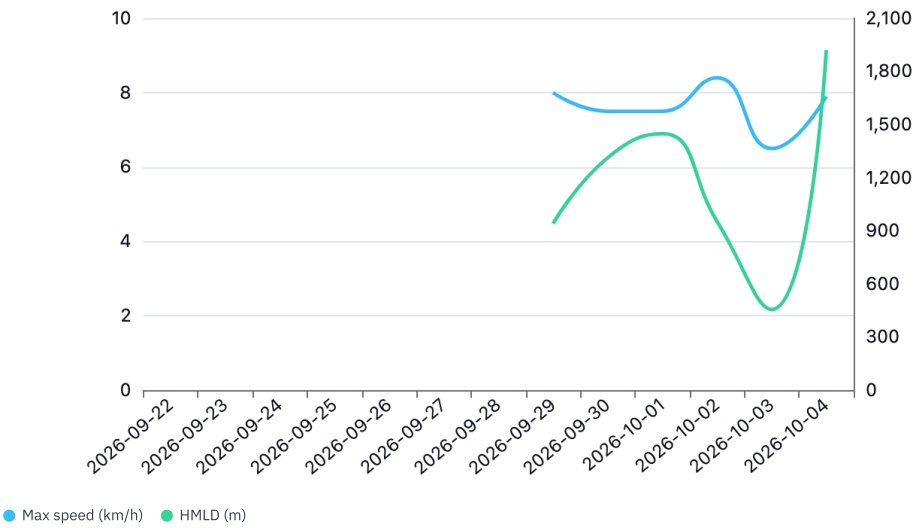
SAMBI LOKONGA – DAILY GPS & WELLNESS PROFILE

Date	Distance total (m)	HMLD (m)	Max speed (km/h)	Accelerations (relative)	Decelerations (relative)	Body Soreness Rating	Performance Score
2026-09-22	5	931	7.6	72	35	10	49
2026-09-29	7.4	941	8	88	48	10	52
2026-09-30	6.5	1.3	7.5	66	48	–	52
2026-10-01	7.8	1.4	7.5	55	31	20	53
2026-10-02	6.9	954	8.4	71	37	30	–
2026-10-03	4	456	6.5	25	15	–	–
2026-10-04	10	1.9	7.9	49	44	–	–
2026-09-28	–	–	–	–	–	–	52
2026-09-27	–	–	–	–	–	–	50
2026-09-26	–	–	–	–	–	–	49
2026-09-25	–	–	–	–	–	–	48
2026-09-24	–	–	–	–	–	–	51
2026-09-23	–	–	–	–	–	–	49

SAMBI LOKONGA – INJURY RISK & PERFORMANCE TREND



SAMBI LOKONGA – HMLD & MAX SPEED EVOLUTION



- Sambi Lokonga shows persistently elevated injury risk (peak 56, mean 50.1) despite moderate GPS load. Max speed declining from 7.6 to 7.2 km/h over period. Accel/decel ratio highly imbalanced (mean 2.06:1, squad mean ~1.3:1), indicating eccentric loading deficit. Body soreness ratings low (10-30) do not reflect underlying risk — disconnect suggests poor subjective awareness or chronic adaptation.

CASE STUDY: YUSSUF POULSEN – VOLUME-RISK MISMATCH

2026-09-22 → 2026-10-05

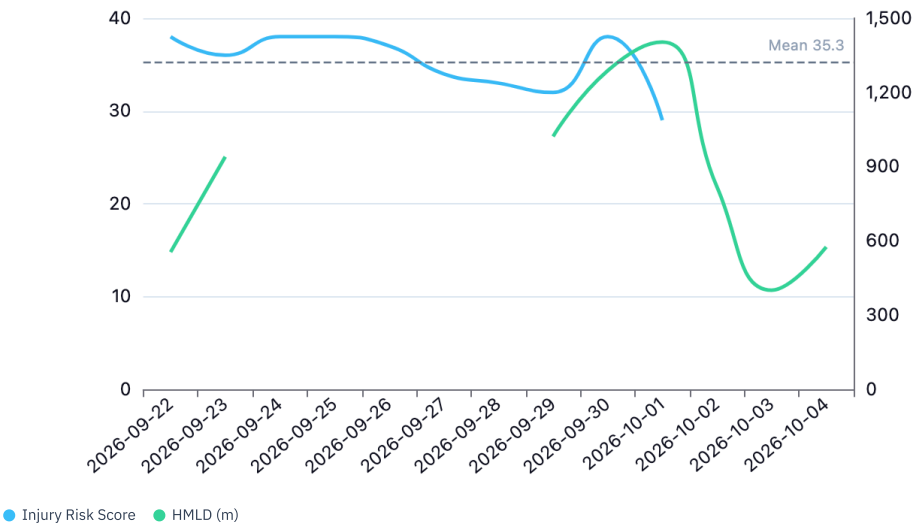
YP

Yussuf Poulsen
STRIKER

POULSEN – DAILY GPS & WELLNESS PROFILE

Date	Distance total (m)	HMLD (m)	High speed running (absolute)	Max speed (km/h)	Body Soreness Rating	Performance Score
2026-09-22	3.6	555	33	7.2	20	49
2026-09-23	5.5	940	599	8.4	10	48
2026-09-29	6.9	1	72	6.8	10	52
2026-09-30	6.3	1.3	330	7.4	–	51
2026-10-01	7.2	1.4	568	7.9	20	53
2026-10-02	5.3	818	134	9	10	–
2026-10-03	3.5	401	126	7.7	–	–
2026-10-04	2.7	576	193	8.8	–	–
2026-09-24	–	–	–	–	20	51
2026-09-25	–	–	–	–	10	52
2026-09-26	–	–	–	–	10	51
2026-09-27	–	–	–	–	10	50
2026-09-28	–	–	–	–	10	52

POULSEN – INJURY RISK VS HMLD



- Poulsen: injury risk 35.3 with below-squad-mean HMLD (875 m/session vs squad ~1,100 m). DSL extremely low (mean 75.9 vs squad ~180), suggesting underload or modified training. High sprint volume (mean 104.5 m) but low mechanical load indicates incomplete neuromuscular preparation. Risk driven by fitness-fatigue imbalance, not acute overload.

CASE STUDY: MIRO MUHEIM – HIGH VOLUME, LOW RISK

2026-09-22 → 2026-10-05

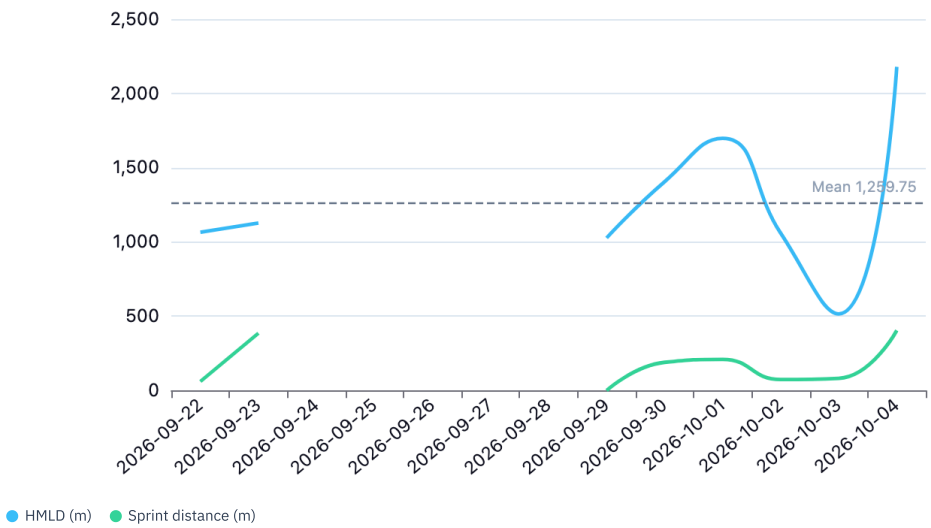
MM

Miro Muheim
FULL BACK

MUHEIM – DAILY GPS & WELLNESS PROFILE

Date	Distance total (m)	HMLD (m)	High speed running (absolute)	Sprint distance (m)	Max speed (km/h)	Performance Score
2026-09-22	5.6	1.1	322	61	8.4	50
2026-09-23	5.1	1.1	619	386	8.2	48
2026-09-29	7.8	1	78	0	7	50
2026-09-30	6.5	1.4	576	190	8.8	50
2026-10-01	9.1	1.7	598	210	8.8	52
2026-10-02	6.9	1.1	236	74	8.8	–
2026-10-03	4	516	240	82	8.1	–
2026-10-04	9.3	2.2	970	405	9	–
2026-09-28	–	–	–	–	–	50
2026-09-27	–	–	–	–	–	49
2026-09-26	–	–	–	–	–	50
2026-09-25	–	–	–	–	–	49
2026-09-24	–	–	–	–	–	50

MUHEIM – HMLD & SPRINT DISTANCE EVOLUTION



- Muheim: highest HMLD exposure (10,078 m total, mean 1,260 m/session) yet injury risk only 17.6. Sprint volume 1,408 m (squad leader). Balanced accel/decel profile (77:70 ratio ~1.1:1). Max speed stable 8.4–9.0 km/h. Demonstrates robust

tolerance to high mechanical load — reference profile for squad capacity benchmarking.

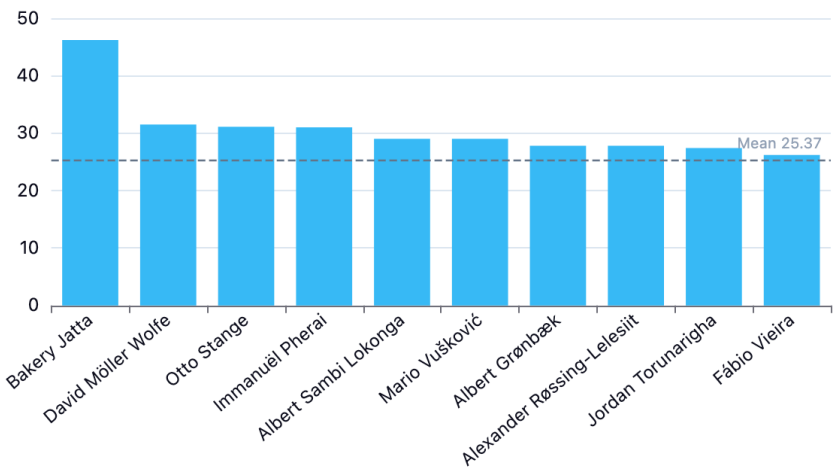
CROSS-DOMAIN ANALYSIS: GPS × WELLNESS × PERFORMANCE

2026-09-22 → 2026-10-05

WELLNESS & RISK SUMMARY – ACTIVE PLAYERS (DISPLAY PERIOD)

Player	N	Distance total (mean)	Distance total (sum)	Distance total (max)	HMLD (mean)	HMLD (sum)	HMLD (max)	High speed running (absolute) (mean)	High speed running (absolute) (sum)	High speed running (absolute) (max)	Sprint distance (mean)	S d (s
Miro Muheim	13	6,791.25	54.3	9.3	1,259.75	10.1	2.2	454.9	3.6	970	176	1
Yussuf Poulsen	13	5,124.38	41	7.2	875.4	7	1.4	256.9	2.1	599	104.5	8
Kofi Amoako	11	5,790.29	40.5	8.3	980.1	6.9	1.7	337.3	2.4	896	110.7	7
Martin Adeline	5	9,508.6	47.5	12.5	1,985.2	9.9	3	571.2	2.9	1.2	152.4	7
Fábio Vieira	11	6,299.67	37.8	7.5	977.8	5.9	1.4	306.5	1.8	547	122	7
Shafiq Nandja	11	7,784.6	38.9	10.6	1,383.4	6.9	1.7	446.2	2.2	691	130.2	6
Jordan Torunarigha	13	5,074.5	30.4	7.2	712.5	4.3	1.1	211.7	1.3	497	90	5
Bilal Nadir	8	5,076.13	40.6	9	736.8	5.9	1.4	198.5	1.6	433	57.5	4
Bakery Jatta	12	6,105.25	24.4	9.4	982	3.9	1.6	291.5	1.2	504	96.8	3
Albert Sambi Lokonga	13	6,794.71	47.6	10	1,137.71	8	1.9	329.7	2.3	652	54.7	3
Sebastiaan Bornauw	6	6,903.17	41.4	9.7	1,027.67	6.2	1.7	218.3	1.3	471	61.3	3
Nicolai Remberg	13	8,477.17	50.9	11.6	1,432.33	8.6	2.3	299.2	1.8	552	55.2	3

TRAINING RPE DISTRIBUTION – DISPLAY PERIOD



MULTI-DOMAIN RISK PROFILE – KEY PLAYERS

Player	HMLD (mean)	DSL (mean)	Max speed (max)	Performance Score Mean
Alexander Røssing-Lelesiit	1,394.8	188	8.8	52.9
Mario Vušković	880.6	181.4	9.2	51.1
Bilal Nadir	736.8	145	9.2	–
Bakery Jatta	982	167.9	8.7	55.2
Yussuf Poulsen	875.4	75.9	9	50.9
Kofi Amoako	980.1	219.6	9.4	–
Miro Muheim	1,259.75	191.2	9	49.8
Albert Sambi Lokonga	1,137.71	215.9	8.4	50.5
Fábio Vieira	977.8	189.2	8.4	49.7
Jordan Torunarigha	712.5	145.9	9.1	48.9
Nicolás Capaldo	671.3	178.3	9.5	50.5
Nicolai Remberg	1,432.33	352	9.2	69.3
Martin Adeline	1,985.2	386.4	9.3	–
Sebastiaan Bornauw	1,027.67	195.1	9.7	–
Terem Moffi	324.2	79.9	9	–
Shafiq Nandja	1,383.4	168.1	10.2	–
Albert Grønbæk	164	139.3	6.4	54.2
Otto Stange	1.6	108	8.8	51.2
David Möller Wolfe	–	–	–	–
Patson Daka	–	–	–	–
Sebastiaan Bornauw	–	–	–	–
Warmed Omari	–	–	–	50.9
Fabio Baldé	–	–	–	51
Jean-Luc Dompé	–	–	–	51
Damion Downs	–	–	–	51
Daniel Elfadli	–	–	–	51
Robert Glatzel	–	–	–	51
Giorgi Gocholeishvili	–	–	–	51
Noah Katterbach	–	–	–	51
Ransford Königsdörffer	–	–	–	51
Omar Megeed	–	–	–	51
William Mikelbrencis	–	–	–	51
Philip Otele	–	–	–	51
Immanuël Pherai	–	–	–	50.5
Rayan Philippe	–	–	–	51
Emir Sahiti	–	–	–	51
Luka Vušković	–	–	–	51

- Weak correlation between subjective soreness and injury risk (r = -0.15, n=56) suggests athletes underreport or normalize chronic fatigue. Sambi Lokonga reports low soreness (10-30) despite risk 50+. Poulsen similar pattern. Recommendation: prioritize objective GPS derivatives (accel/decel imbalance, max speed decline) over self-report for injury flagging.

TECHNICAL RECOMMENDATIONS

2026-09-22 → 2026-10-05

- Sambi Lokonga: Reduce eccentric loading volume. Accel:decel ratio 2.06:1 (target <1.5:1). Prescribe dedicated deceleration drills. Monitor max speed weekly — current decline 7.6→7.2 km/h flags neuromuscular fatigue. 2. Poulsen: Increase DSL exposure gradually (current 75.9 m, target 150+ m). Low mechanical load despite sprint volume indicates incomplete conditioning. Risk driven by fitness gap, not overload. 3. Sq...

SUMMARY & KEY FINDINGS

2026-09-22 → 2026-10-05

- Period analysis (22 Sep – 5 Oct) reveals three distinct injury risk profiles: (1) High-risk despite moderate load (Sambi Lokonga, Poulsen) — driven by mechanical imbalance and neuromuscular decline; (2) Low-risk high-volume (Muheim, Remberg) — robust tolerance benchmarks; (3) Elevated risk without GPS data (Königsdörffer, Glatzel) — non-training factors. Advanced derivatives (HMLD/min, accel/decel ratio, max speed...

PERFORMANCE SCORE VS INJURY RISK – SQUAD COMPARISON

